

Benchmark #4

1MeV Xe on UO2

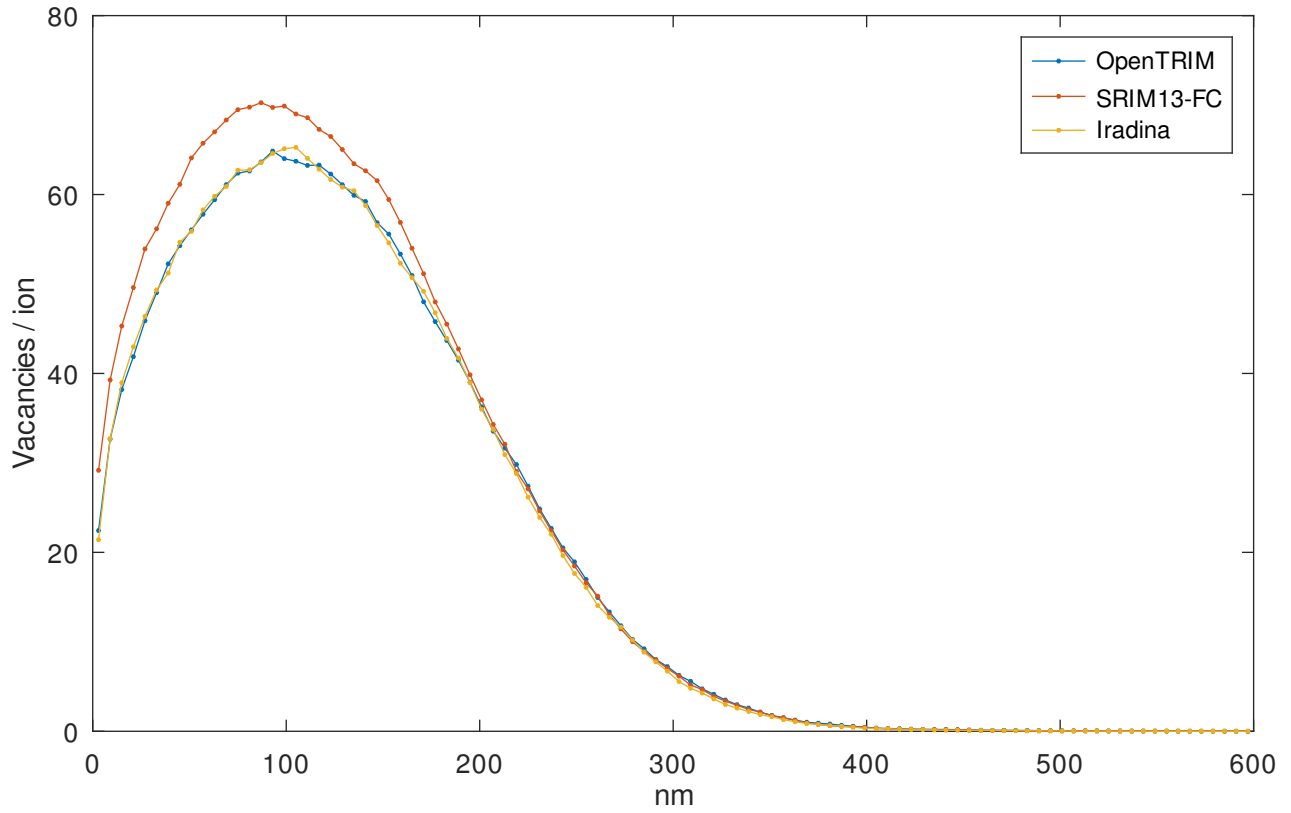
Ion energy $E_0 = 1\text{e}+06$ eV

Target depth = 600 nm

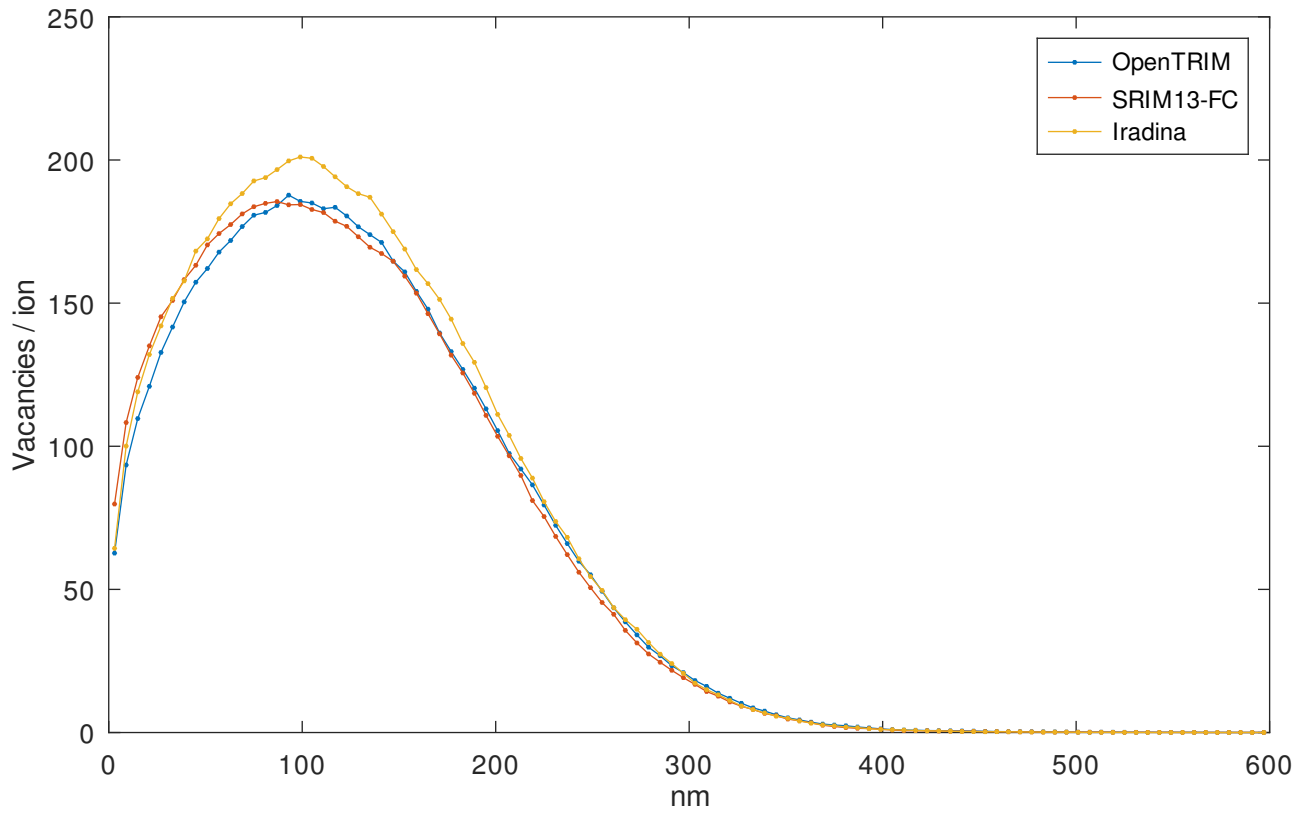
Summary Table

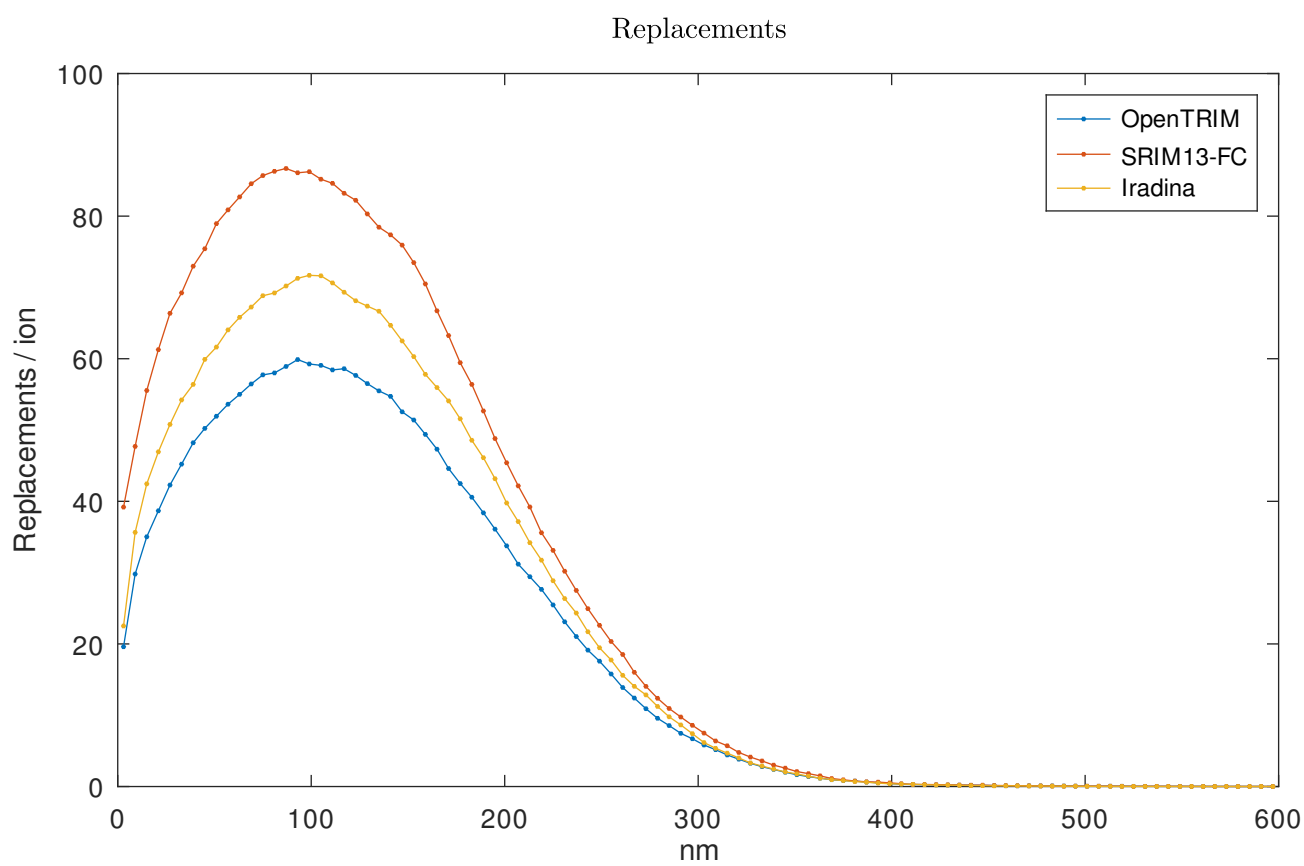
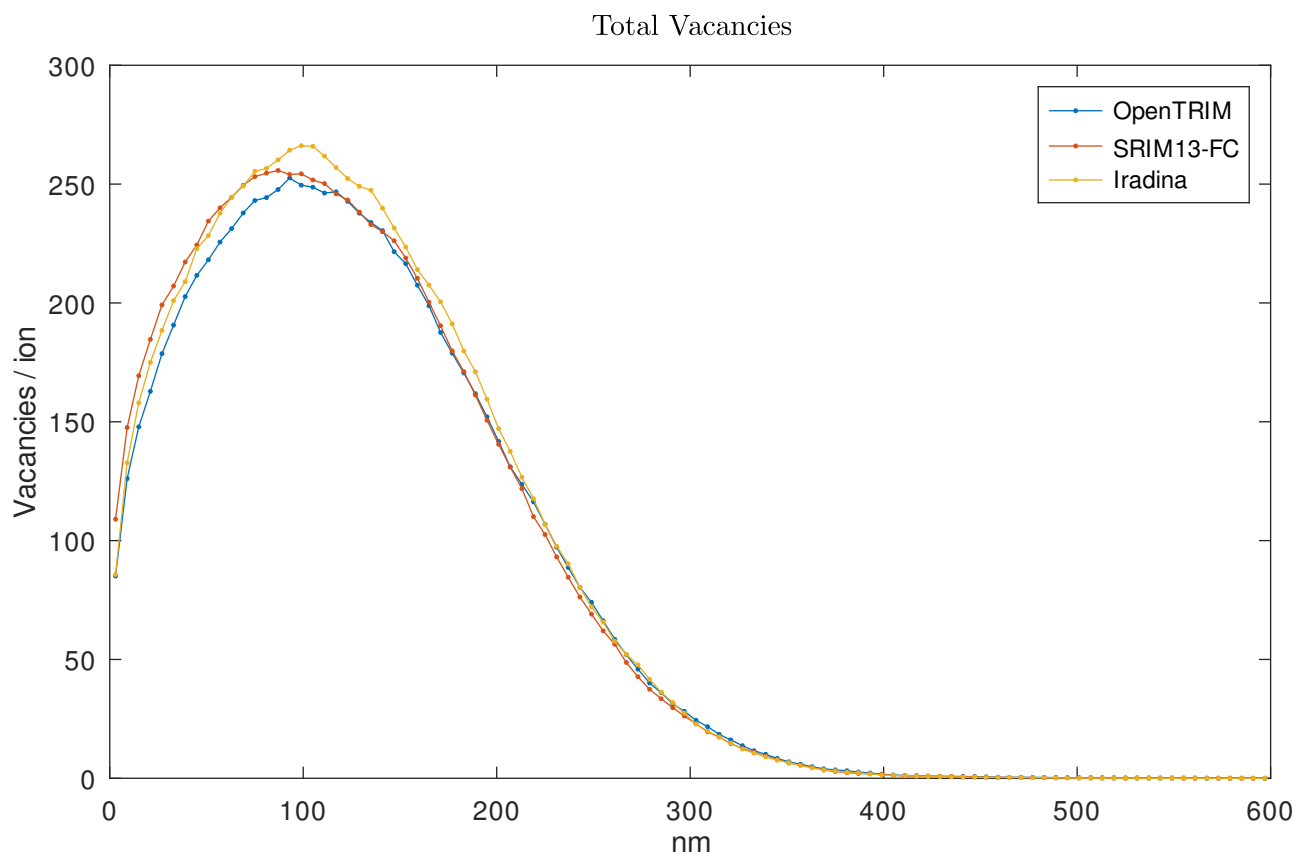
Quantity	OpenTRIM	SRIM13-FC	Iradina
V(U)	2.14e+03	2.31e+03	2.12e+03
V(O)	6.19e+03	6.21e+03	6.55e+03
V(tot)	8.33e+03	8.52e+03	8.68e+03
R(tot)	1.98e+03	2.85e+03	2.34e+03
I(Xe)	0.991	0.986	0.988
$\Delta E_e(\text{Xe})/E_0$	0.162	0.16	0
$\Delta E_e(r)/E_0$	0.332	0.346	0
$\Delta E_e(\text{tot})/E_0$	0.494	0.507	0.534
$\Delta E_n(\text{Xe})/E_0$	0.00252	0.00241	0
$\Delta E_n(r)/E_0$	0.497	0.452	0
$\Delta E_n(\text{tot})/E_0$	0.499	0.455	0.458
$1 - (\Delta E_e + \Delta E_n)/E_0$	0.00691	0.0388	0.00728

Vacancies of U in Uranium oxide

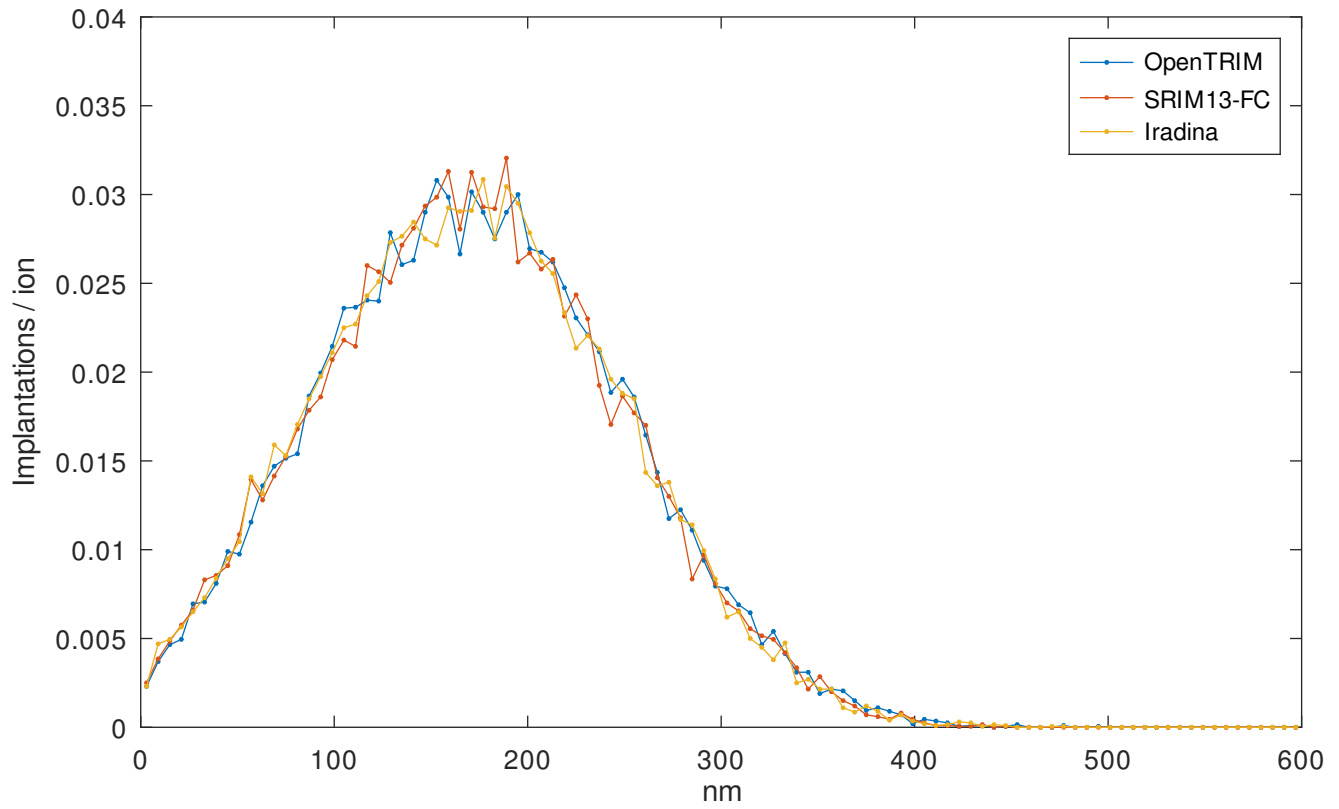


Vacancies of O in Uranium oxide

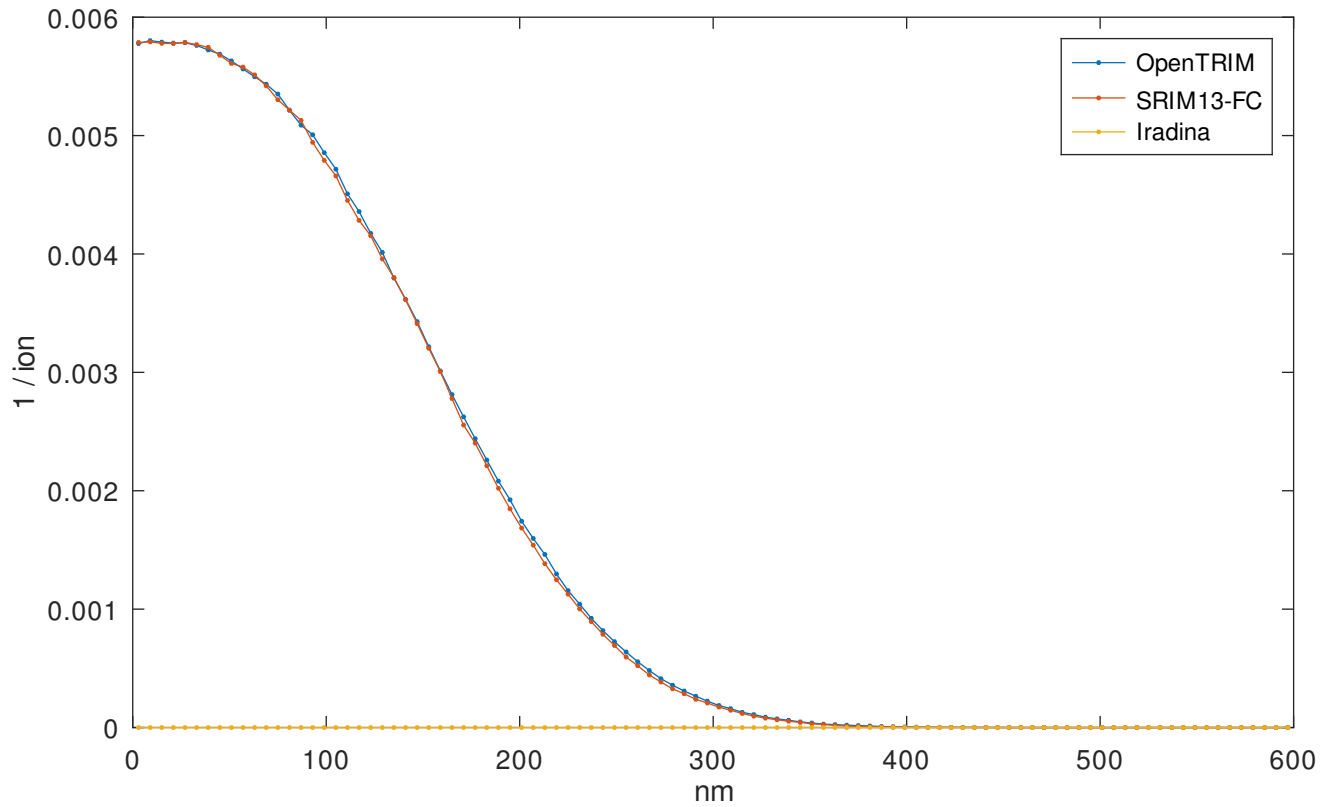




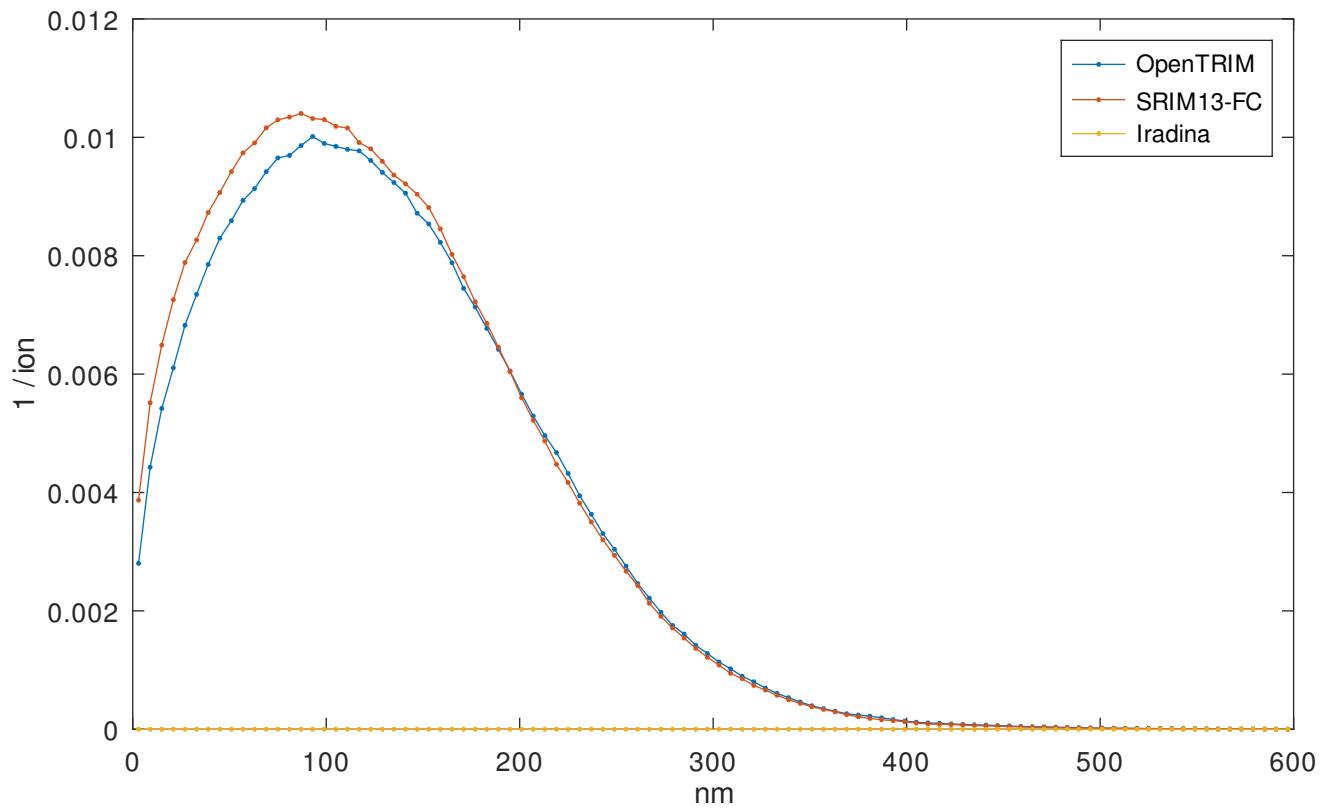
Implanted Xe ion



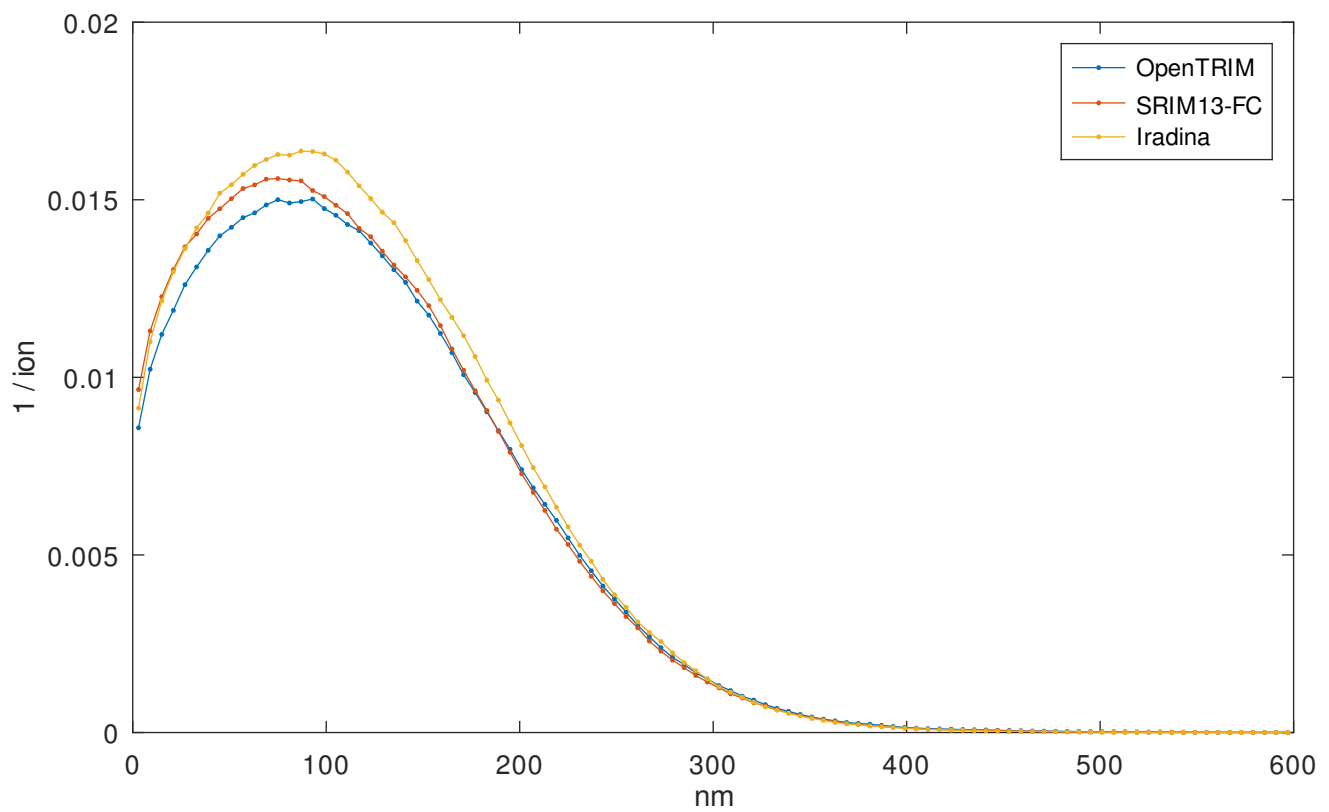
Electronic energy loss fraction $\Delta E_e/E_0$ by Xe ion



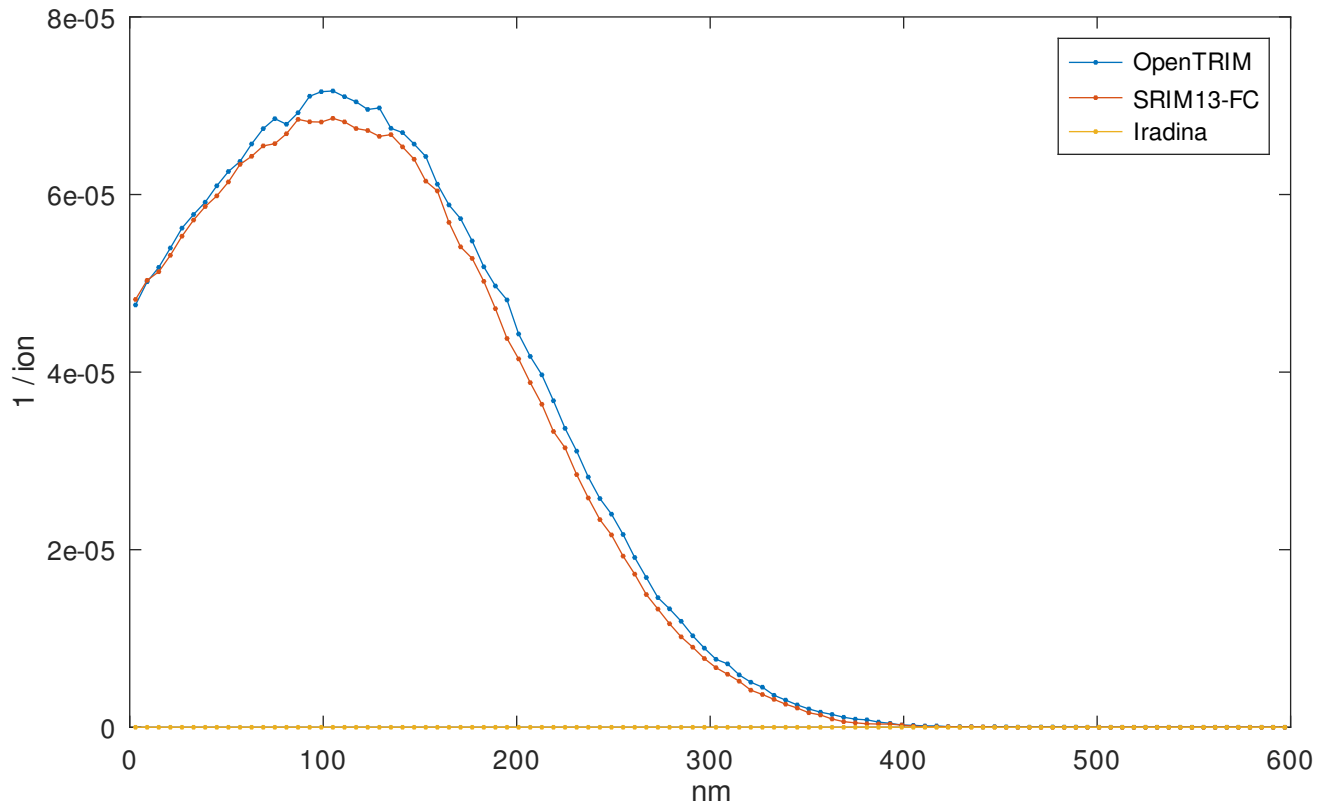
Electronic energy loss fraction $\Delta E_e/E_0$ by recoils



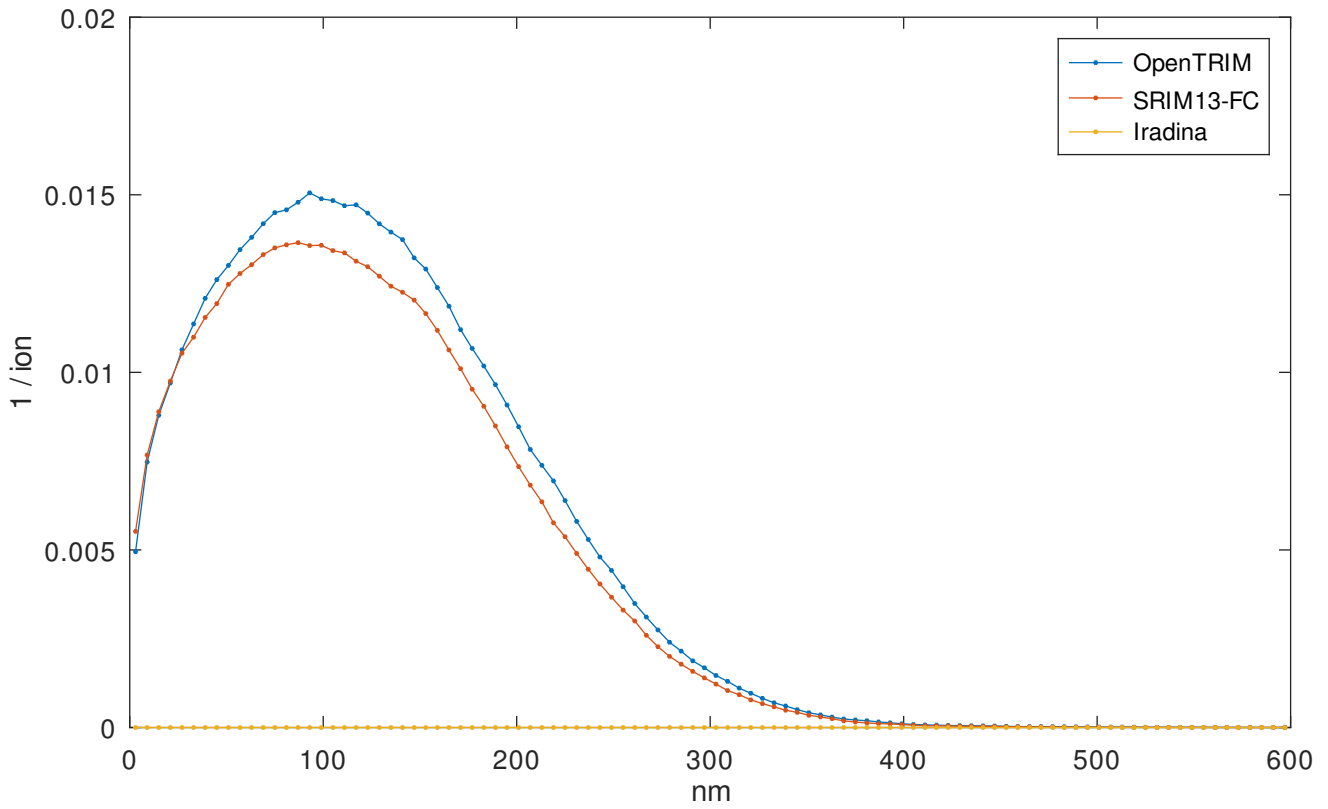
Total Electronic energy loss fraction $\Delta E_e/E_0$



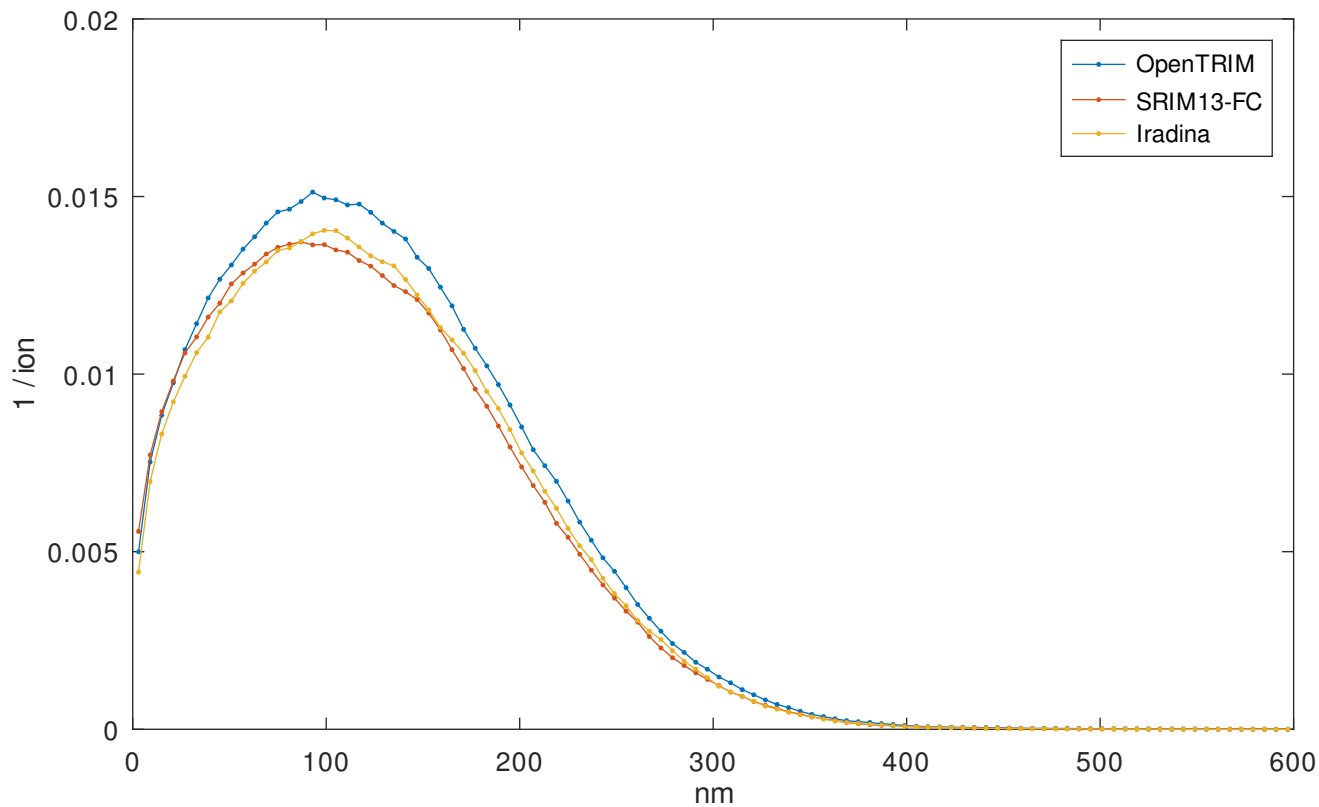
Nuclear energy loss fraction $\Delta E_n/E_0$ by Xe ion



Nuclear energy loss fraction $\Delta E_n/E_0$ by recoils



Total Nuclear energy loss fraction $\Delta E_n/E_0$



Total fractional energy loss $(\Delta E_e + \Delta E_n)/E_0$

